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PROFILE

Machine Learning Engineer with 3 years of experience building production ML and NLP systems for financial services clients. I specialise in large language models, transformers and gradient boosted models for structured information extraction from unstructured financial text. At Accenture I own enterprise GenAI systems end to end, from research and experimentation through production deployment. This covers real time NLP pipelines, RAG over dense vector databases, and the evaluation frameworks and annotation tooling that gate every release, including LLM as a judge scoring, error taxonomy analysis and hallucination tracking. My research foundation comes from an ML internship at IIT Kharagpur, where I applied computer vision to a wildlife conservation programme, a practical grounding for sustainability focused ML, alongside a GATE result in the top 2% nationally. I work well with domain experts, product managers and business stakeholders, and I am good at turning model failure modes into measurable improvements in production reliability.

TECHNICAL SKILLS

Machine Learning and NLP: Large language models (GPT, Claude), transformers, gradient boosted decision tree ensembles, PyTorch, scikit learn, NLP pipelines, structured information extraction, prompt engineering, fine tuning (LoRA, QLoRA, DPO), text classification, embeddings

Retrieval and LLM Systems: RAG architecture, dense vector databases, semantic search, chunking and vectorisation strategies, LangGraph, ReAct, agent orchestration, function calling, guardrails

Evaluation and Model Quality: LLM as a judge evaluation frameworks, human review and annotation workflows, hallucination and error rate tracking, failure mode and error taxonomy analysis, automated regression testing, model selection studies comparing RAG against fine tuning

Engineering and MLOps: Python in production, model serving, data pipelines, Apache Airflow, Docker, Kubernetes, CI/CD, AWS, Snowflake, REST APIs, SAP S/4HANA and BTP integration, production monitoring

Fundamentals: Data structures and algorithms, statistics, time series modelling, SQL

WORK EXPERIENCE

Accenture, AI/ML Engineer (Forward Deployed)

Jul 2023 to Present

Kolkata and Bengaluru, India · Enterprise GenAI for financial services and pharmaceutical clients

VERTEX, an enterprise tax automation engine (LangGraph agents)

- Own a production LLM system end to end, from research and experimentation through deployment. The system performs **structured information extraction** from millions of unstructured financial documents into tax classifications, orchestrated with **LangGraph** and ReAct agent patterns and integrated with SAP HANA and downstream REST APIs.
- Designed the **evaluation framework** that gates every release. GPT and Claude outputs are scored with LLM as a judge against compliance reviewed reference cases, and recurring error types become programmatic guardrails and prompt refinements.
- Built the **annotation and human review tooling**. Low confidence cases route to expert reviewers, and their feedback drives measurable, continuous improvement of the RAG retrieval pipeline.

Invoice Financial Review, an LLM evaluation and MLOps platform

Sole technical owner of a General Ledger and tax classification platform over structured and unstructured invoice data, with **evaluation pipelines and automated regression tests** running on every model iteration.

- Drove model selection decisions from evaluation evidence, comparing GPT against Claude and RAG against fine tuning, and explained the tradeoffs clearly to engineering peers as well as business stakeholders.
- Traced model performance regressions back to latent data quality issues in the raw invoice feed and fixed them at source.

Audit Mail Response System, real time NLP automation

- Built a **real time NLP pipeline using LLMs** over high volumes of unstructured audit correspondence, integrating SAP S/4HANA financial data and the Microsoft Graph API via REST.
- Wrote Python evaluation suites that tracked hallucination and intent classification error rates in production, and reworked the architecture mid delivery when new compliance constraints arrived, without slipping the timeline.

SAP Enterprise Platform and Cloud Engineering

- Built and deployed production data pipelines and cloud native ML infrastructure on SAP BTP and AWS for pharmaceutical and financial regulatory workflows, with CI/CD, Docker containerisation and production monitoring.

IIT Kharagpur, Machine Learning Research Intern

Applied ML research for a wildlife conservation programme, sustainability focused ML in practice

- Built and evaluated image classification models for conservation field research, covering Python data pipelines, rigorous evaluation metrics and failure mode analysis on edge cases. The system was deployed within a conservation effort that reported up to a 50% reduction in poaching in the study area.

SELECTED PROJECTS

Jane Street Market Prediction, Kaggle

PyTorch · gradient boosted ensembles

- Modelled high dimensional, noisy financial market data with a PyTorch MLP combined with a **gradient boosted decision tree ensemble**, designing the cross validation carefully to avoid temporal leakage in financial time series.

AI Investment Research Desk, an agentic research system Python · LangGraph · OpenAI / Anthropic

- Designed a system of five specialised agents coordinating financial research, analysis and risk assessment through a central orchestrator, with a Monte Carlo simulation engine for scenario modelling and live market data via function calling. Published a technical article on the design.

EDUCATION

B.Tech, Electrical Engineering, Institute of Engineering and Management, Kolkata 2018 to 2022

- DGPA 9.08 out of 10, with graduate level research experience through the IIT Kharagpur internship above.
- GATE, the national graduate engineering examination: ranked in the **top 2% nationally**.

PUBLICATIONS AND AWARDS

- Prediction-of-Pediatric-Pneumonia-in-Chest-X-Rays-using-Deep-Learning, Abhirup Majumder (Manuscript in preparation) (2025)
- SYNCHRONIZATION OF PV SYSTEM WITH ELECTRICAL GRID, DOI: 10.13140/RG.2.2.11847.92329 Abhirup Majumder, Pritam Sanyal, Aman Raj, Saikat Banerjee, Soumi Sinha, Subhasis Guha (2023)
- AUTOMATIC DUAL-AXIS SOLAR TRACKER WITH SOLAR PANEL (Published in WJERT Journal) wjert, 2020, Vol. 6, Issue 5, 286-294. ISSN 2454-695X Arjun Dutta, Abhirup Majumder, Akash Das (2020)
- ISHRAE All India National Student Design Competition: 3rd place nationally for an IoT smart climate control system.

References available upon request.